WATER HYDRAULIC MOTORS / PRODUCT INFORMATION

M3 Water Hydraulic Motor

Evaluate the M3 at the required starting torque and running speed. Use its displacement and performance data to confirm flow, pressure difference and drive behaviour; include case-drain limits, stopping and shaft-load requirements.

The M3 motor excels in compact installations where oil contamination would compromise safety, product quality, or environmental compliance. At 0.8 kg (1.76 lb) it delivers reliable torque in food processing, marine, and wastewater applications.



M3 Water Hydraulic Motor

At a glance

Parameter / value	Parameter / value
Displacement 3.0 cc/rev (0.183 in³/rev)	Max Pressure 160 bar (2,321 psi) continuous
Peak Efficiency 86 %	Speed Range 1000-3000 RPM
Max Torque 7.1 Nm (5.24 lbf·ft)	Max Power 2.2 kW (2.95 hp)

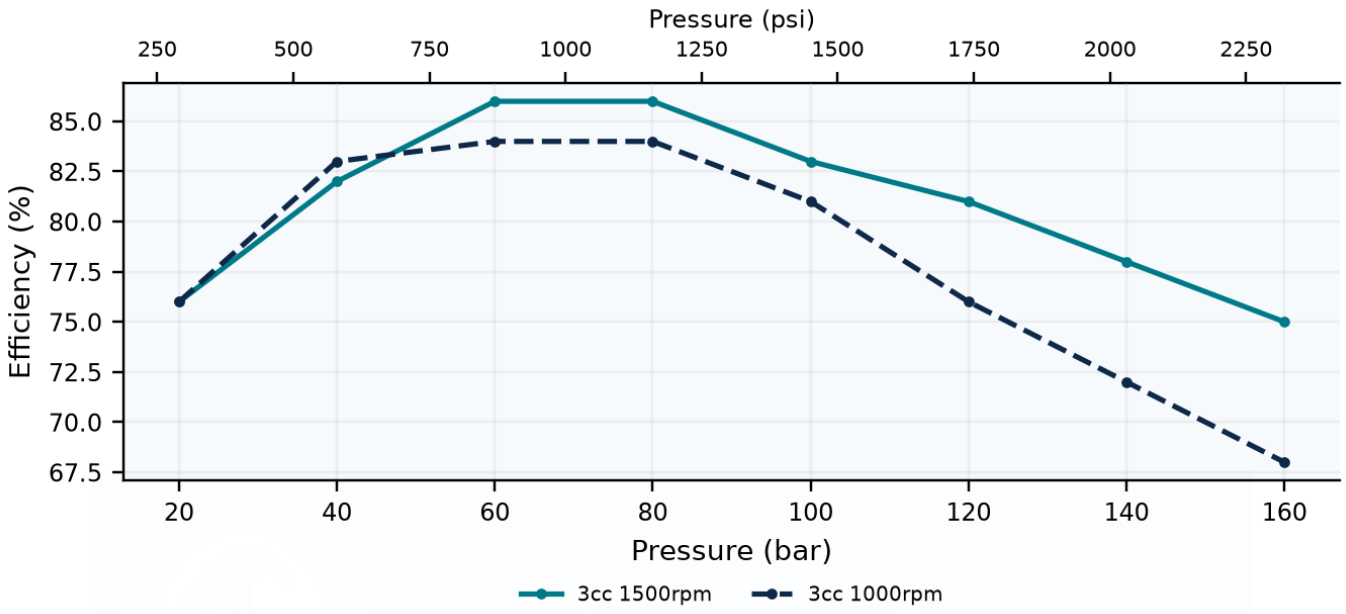
Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

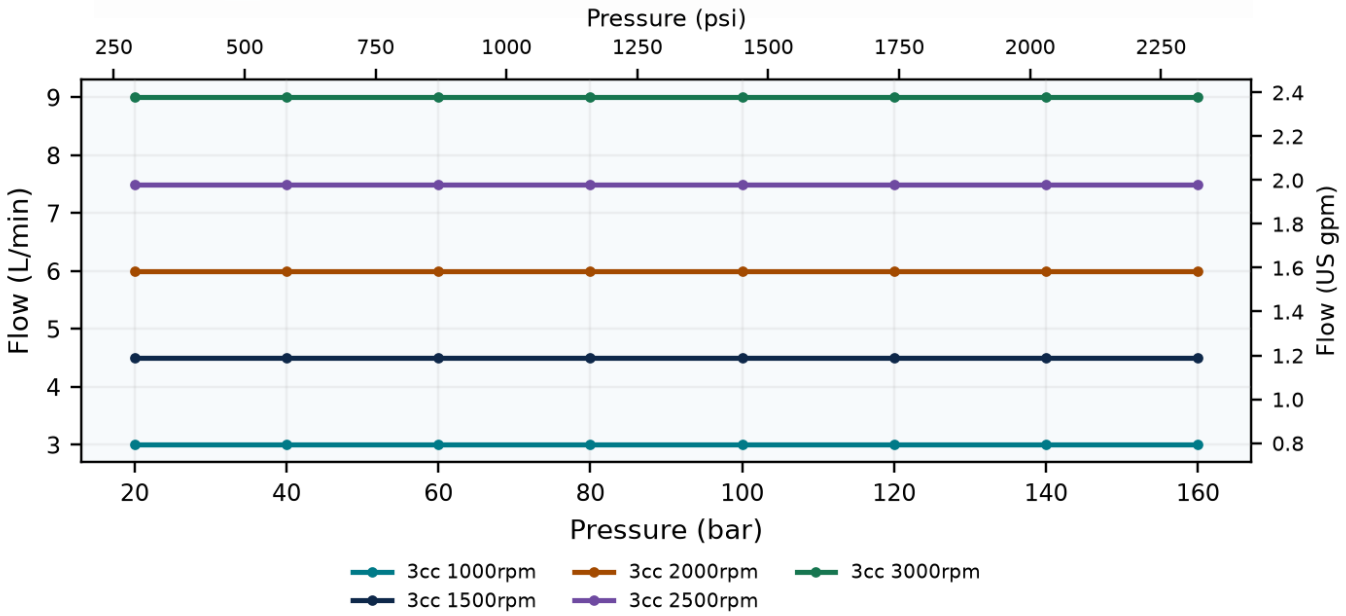
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

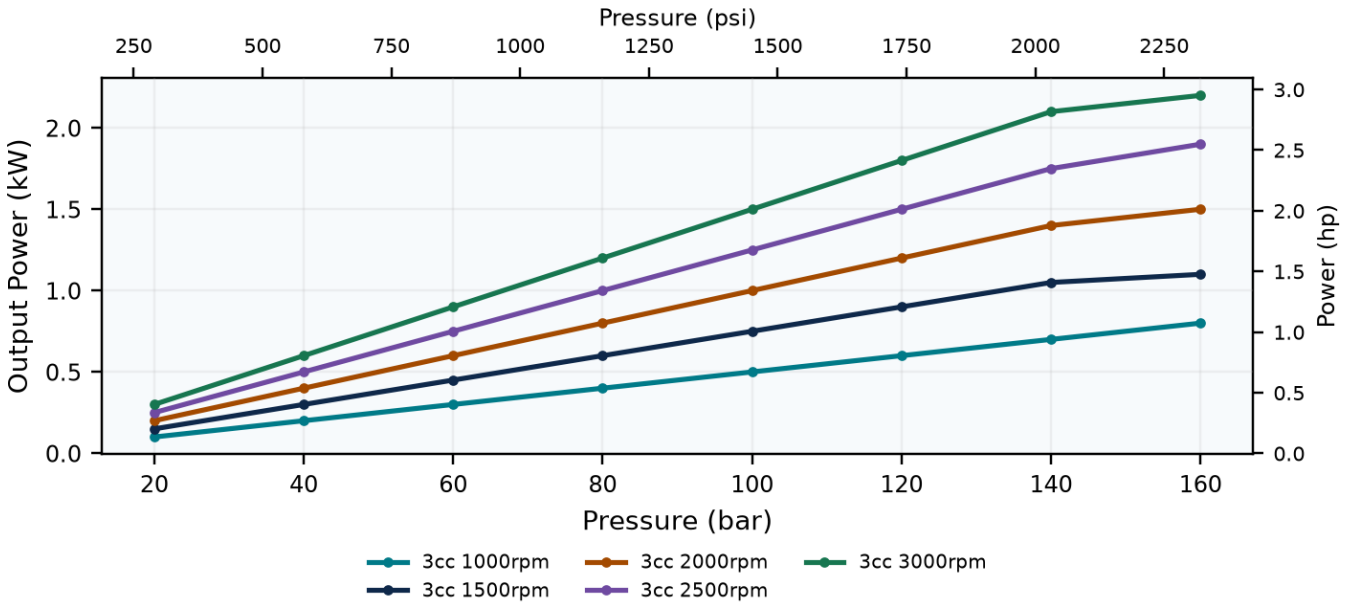
Efficiency (%) vs Pressure (bar)



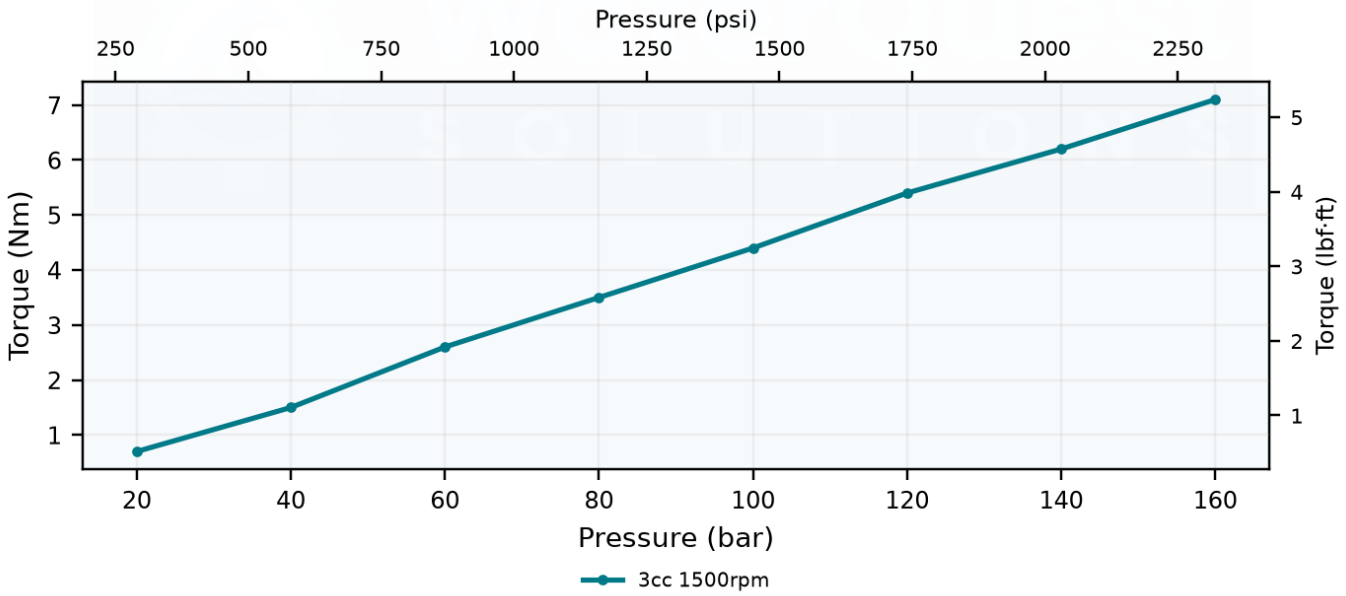
Flow (L/min) vs Pressure (bar)



Output Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)

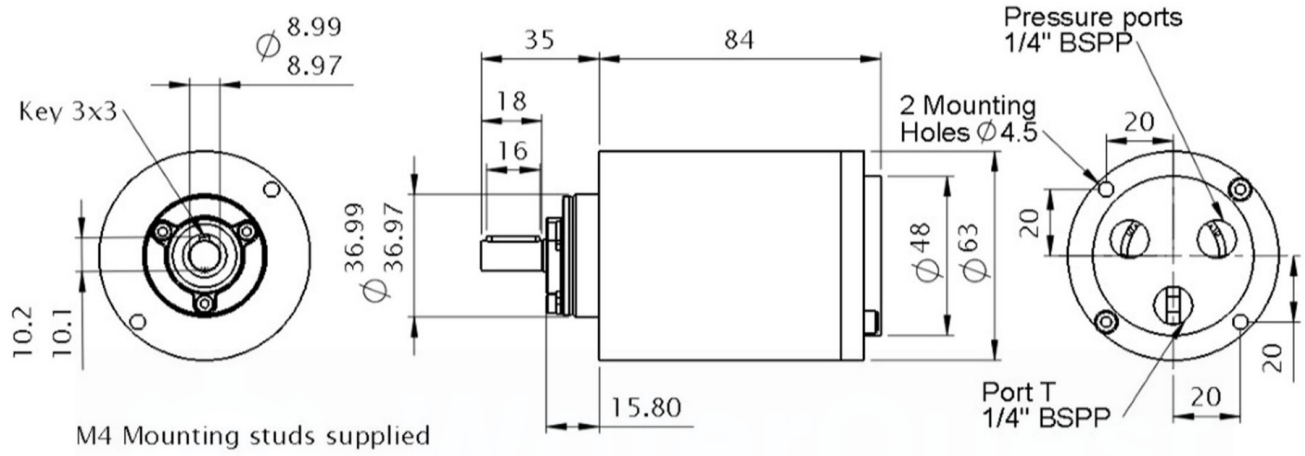


Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

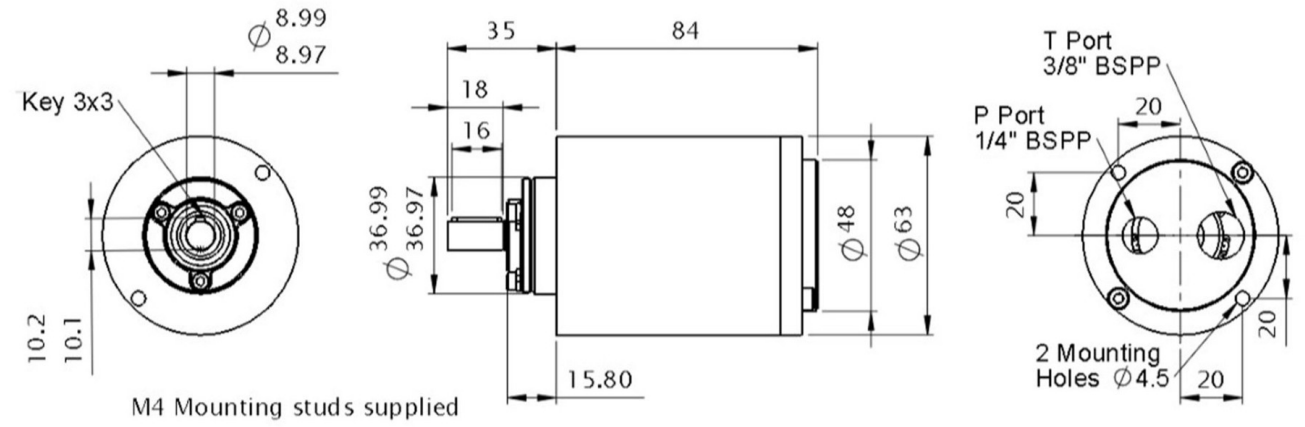
MB160-3.0W Technical Drawing

MB160-3.0W



MA160-3.0W Technical Drawing

MA160-3.0W & MC160-3.0W



Application and specification

Select the wetted material and seal package for the stated chemistry, salinity, temperature and exposure. Standard stainless construction is not a universal seawater approval; confirm the exact configuration with WaterQuest.

Removing hydraulic oil from this circuit removes that oil inventory from a potential release. Water quality, additives, other machine fluids and local discharge requirements still need review for the application.

86% peak efficiency maintained across 60-80 bar (870-1,160 psi) range reduces energy consumption and heat generation versus oil motors of equivalent displacement.

Specify the working-water chemistry and external exposure, then confirm the permitted material and seal package. Seawater operation requires explicit approval for the configuration; it cannot be inferred from standard stainless construction.

84mm (3.31 in) body length enables installation in space-constrained machinery where larger motors cannot fit, critical in food processing retrofits.

MB series provides reversible rotation for winch and positioning systems without valve manifold complexity.